Reg. No. | | | | | | | | | | |

[14AE662]

SIXTH SEMESTER B.Tech UNIVERSITY EXAMINATION, JUNE–2018

OPERATIONS RESEARCH

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE**full Questions selecting at least two questions from each part.
2. Missing data may be suitably assumed.
3. Statistical tables will be provided.

PART-A

1. a. A company manufactures two models of Laptops Model J and Model K which fetch net profits of Rs 5000 per unit and Rs 7000 per unit respectively. Each unit of model J takes 4 hours of assembly time and 2 hours of testing time while each unit of model K requires 3 hours for assembly and 1 hour for testing. In any given month, the total number of hours available for assembly is 210 hours and that for inspection is 90 hours. Company policy is to determine the production quantity of each product in such a way as to maximize the total profit. Formulate this as linear programming problem. **04**

b. Define the following terms: **05**

- **i.** Basic feasible solution
- **ii.** Degenerate solution
- **iii.** Feasible solution
- **iv.** Infeasible solution
- **v.** Unbounded solution

c. Solve the following linear programming problem by graphical method: **05**

Minimize, $Z = 20x_1 + 10x_2$

subject to the constraints $x_1 + 2x_2 \leq 40,$

$$3x_1 + x_2 \geq 30,$$
$$4x_1 + 3x_2 \geq 60,$$
$$x_1, x_2 \geq 0$$

2. a. Solve the following linear programming problem by simplex method: **07**

Maximize $Z = 2x_1 + 2x_2 + 4x_3$

Subject to the constraints$2x_1 + 3x_2 + x_3 \leq 240,$

$$x_1 + x_2 + 3x_3 \leq 300,$$
$$x_1 + 3x_2 + x_3 \leq 300$$
$$x_1, x_2, x_3, \geq 0$$

b. Solve the following linear programming problem by Big M method: **07**

Minimize $Z = x_1 - 2x_2 - 3x_3$

Subject to the constraints $2x_1 + x_2 + 2x_3 = 2,$

$$2x_1 + 3x_2 + 4x_3 = 1$$
$$x_1, x_2, x_3 \geq 0$$

3. a. Apply the principle of duality, to solve the linear programming problem **07**

Maximize $Z = 3x_1 - 2x_2$

Subject to the constraints $x_1 + x_2 \leq 5,$

$$x_1 \leq 4$$
$$1 \leq x_2 \leq 6$$
$$x_1, x_2 \geq 0$$

b. Solve the following linear programming problem by dual simplex method: **07**

Minimize $Z = 2x_1 + x_2$

Subject to the constraints $x_1 + x_2 = 4,$

$$2x_1 - x_2 \geq 3,$$
$$x_1, x_2 \geq 0$$

4. a. The following table shows all the necessary information on the availability of supply to each warehouse, the requirement of each market and unit transportation cost in Rs from each warehouse to each market. Find the optimal solution to the transportation problem by MODI method, given **07**

Factory/ Warehouse	F	G	H	I	Supply
A	6	3	5	4	22
B	5	9	2	7	15
C	5	7	8	6	8
Demand	7	12	17	9	45

b. A department has 5 employees with 5 jobs to be performed. The time in hours each person will take to perform each job is given in the form of matrix. How jobs should be allocated, so as to minimize man hours. **07**

Employees/ Jobs	I	II	III	IV	V
A	10	5	13	15	16
B	3	9	18	13	6
C	10	7	2	2	2
D	7	11	9	7	12
E	7	9	10	4	12

PART-B

5. a. Use Revised simplex method to **07**

Maximize $Z = 3x_1 + 5x_2$

Subject to the constraints

$x_1 \leq 4,$

$x_2 \leq 6,$

$3x_1 + 2x_2 \leq 18,$

$x_1, x_2, \geq 0$

b. Solve the following integer programming problem: **07**

Maximize $Z = x_1 + 2x_2$

Subject to the constraints

$2x_2 \leq 7,$

$x_1 + x_2 \leq 7,$

$2x_1 \leq 11$

$x_1, x_2\ arenonnegativeintegers$

6. a. Define: **04**

i. Saddle point

ii. Value of a game

iii. Max-min Principle

iv. Two person Zero sum game

b. Use Dominance rule to solve the following game: **05**

	B_1	B_2	B_3	B_4
A_1	3	2	4	0
A_2	3	4	2	4
A_3	4	2	4	0
A_4	0	4	0	8

c. Solve the following game graphically whose payoff matrix for player A is **05**

	B_1	B_2	B_3	B_4	B_5
A_1	2	-4	6	-3	5
A_2	-3	4	-4	1	0

7. a. An assembly operation involves the completion of 11 jobs. The job labels, the time required to complete each one and the necessary immediate predecessors of each job are shown in the table: **06**

Job	Job time (days)	Immediate predecessors
A	13	-
B	08	A
C	10	B
D	9	C
E	11	B
F	10	E
G	8	D,F
H	6	E
I	7	H
J	14	G,I
K	18	J

Draw a network diagram for the project and find the critical path and total float of each activity

b. The activities of a project have the following PERT time estimates **08**

Job i-j	Optimistic time (days)	Most likely time	Pessimistic time
1-2	3	6	15
1-6	2	5	14
2-3	6	12	30
2-4	2	5	8
3-5	5	11	17
4-5	3	6	15
6-7	3	9	27
5-8	1	4	7
7-8	4	19	28

i. Draw the network diagram and determine critical path

ii. Find the project completion time. And variance of the critical path.

iii. What is the approximate probability that the jobs on the critical will be completed in 41 days?

iv. Find the project duration at 90% confidence.

8. a. Explain Kendall's notation for queuing model. Give description of M/M/1:$(\infty/FIFO)$ model. **07**

b. The arrival of customers at a banking counter follows Poisson distribution with a mean of 45 per hour. The service rate of the counter clerk also follows Poisson distribution with a mean of 60 per hour. **07**

i. Determine the steady state performance statistics L_s, L_q, W_q, W_s.

ii. What is the probability of five customers in the system?